

TEMPORAL DECOUPLING

Toward a Unified Theory

Time · Gravity · Quantum Mechanics · The Timeless Sea

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Version 5 · 2026

Comprising Two Frameworks

Timeless Sea Cosmology (TSC) · Accumulative Emergent Time Theory (AETT)

This document presents an original conceptual theoretical framework first articulated in late 2025. It does not yet contain complete mathematical derivations. It is intended as a structured record of original ideas and a foundation for future formalisation and experimental investigation.

Available at: [zenodo.org/\[DOI\]](#) · [github.com/\[repository\]](#)
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Introduction

This framework proposes two fundamental reinterpretations: time and gravity are not fundamental properties of reality. Both emerge from a deeper structure: a timeless medium called the **Timeless Sea**, within which our universe formed and continues to exist.

Gravity is not a force. It is the inward pressure of the Timeless Sea acting on our universe from outside. **Time** is not a background stage. It is a local phenomenon that emerges wherever matter accumulates and timeline coupling deepens.

What follows is not a finished theory. It is a framework — built on a single postulate, generating testable predictions, and inviting mathematical formalisation and experimental scrutiny.

A note on v5: This version corrects a conceptual conflation present in earlier drafts. Previous wording sometimes treated gravity as the cause of time dilation or decoherence. The corrected framework makes **matter density ρ** the fundamental causal variable. Local time rate $T(\rho)$ and observed gravity (TSC pressure gradient) are parallel emergent effects of ρ , not cause and effect.

Preface: A Vision

This framework proposes that distance, time, and the speed-of-light barrier are not fundamental — they are properties of our universe that emerged from a deeper timeless medium. If that is true, learning to work with that medium changes everything.

Every limitation of current civilisation — energy scarcity, communication delays, the fragility of quantum computing — is a consequence of being fully coupled to our timeline. Temporal Decoupling describes how to understand that coupling, and ultimately, how to work with it.

If validated, the framework may imply profound technological consequences.

The Architecture of the Framework

Timeless Sea Cosmology (TSC) describes the medium itself — its properties, its relationship to our universe, and how it gives rise to gravity, dark matter, dark energy, and the conditions that made the Big Bang possible.

Accumulative Emergent Time Theory (AETT) describes our universe — how time emerged from matter accumulation and expansion, why it flows in one direction, how it governs quantum coherence, and why quantum particles behave as they do when observed.

Decoupling is the mechanism connecting both frameworks. Particles decouple from the Timeless Sea into our timeline when measured. Qubits lose coherence as they are captured by the timeline. Gravity is the pressure of the Timeless Sea pushing inward on our universe.

A quantum particle is the observable spacetime intersection of a static Timeless Sea structure with a temporally evolving spacetime manifold.

PART ONE: CORE MECHANICS

1. The Core Postulate

There exists a state of reality outside our universe's timeline — the Timeless Sea — in which quantum particles exist in superposition, unconstrained by time or gravity. Our universe formed within this Timeless Sea as a structured event. Time itself is not fundamental but emergent.

The Timeless Sea is not claimed to be the ultimate beginning. There may be structures beyond it, or the question of 'beginning' may have no meaning outside a timeline. The framework does not speculate beyond what is in principle testable from within our universe.

1.1 Properties of the Timeless Sea

- Exists outside the spacetime structure of our universe
- Contains **flow lines** — minimal-energy geodesic structures: oriented, one-dimensional objects with no mass, charge, or spin (these emerge only when a flow line intersects our timeline)
- Exerts **pressure** on our universe from outside
- Distance as we understand it does not exist within it
- Retains information permanently

1.2 Relationship to Existing Physics

The Timeless Sea postulate is consistent with several established or seriously proposed ideas: the Hartle–Hawking no-boundary proposal, Carlo Rovelli’s relational quantum mechanics, Julian Barbour’s work on emergent time, and Erik Verlinde’s entropic gravity.

The Timeless Sea differs from these predecessors in being explicitly a **medium** — not just a mathematical structure or philosophical principle — with physical properties (pressure, coherence volumes, flow lines, information retention) that generate specific, falsifiable predictions.

2. The Emergence of Time

Time is not fundamental. It emerged within our universe as a consequence of matter accumulation and expansion. The time rate follows a power law relationship with local matter density:

$$T(\rho) = \beta \rho^n \quad \text{with } n \approx 0.04 \pm 0.01$$

The exponent $n \approx 0.04$ is constrained simultaneously by three independent observational datasets: void region Hubble constant measurements, the Hubble tension, and the Webb galaxy age excess. This is the first quantitative parameter in the framework derived from data rather than theoretical argument alone.

Causal clarity: Matter density ρ determines $\Gamma(\rho)$ (timeline coupling strength). $\Gamma(\rho)$ determines local time rate $T(\rho)$. Gravity — the observed force — emerges separately from the TSC pressure gradient $\nabla P = 4\pi G\rho$. Gravity does not cause time dilation. Matter density causes both. The observed correlation (denser regions = slower clocks = stronger gravity) is a correlation between parallel emergent effects, not a causal link.

2.1 The Arrow of Time

Time flows in the direction of increasing coupling to the timeline. Entropy increase is the same process as timeline coupling deepening:

$$dS/dt = k \times \Gamma_{\text{coupling}}$$

The thermodynamic and experiential arrows are the same arrow, pointing in the same direction for the same reason.

3. Gravity as Timeless Sea Pressure

Gravity is not a fundamental force mediated by a particle. It is emergent pressure: the inward pressure of the Timeless Sea acting on our universe from outside.

The equations of gravity are identical to those of general relativity because both describe the same observable result. GR describes the geometry of that result. This framework proposes the mechanism underneath it.

3.1 The Bulge, Not a Well

General relativity models gravity as a depression in spacetime. This framework proposes an alternative interpretation with identical mathematics: massive objects create a **bulge outward** into the Timeless Sea rather than a sink inward into spacetime.

Objects near a mass are not falling into a well. They are being pushed toward it by the Timeless Sea pressure differential.

3.2 Why Gravity Is the Weakest Force

Gravity is external TSC pressure. All other forces — electromagnetic, strong, weak — are internal interactions within the timeline. Gravity being orders of magnitude weaker than those forces is therefore expected: it is a residual effect of an external medium acting on our timeline from outside.

3.3 Black Holes as Maximum Contact Points

A black hole is a point of maximum contact between our timeline and the Timeless Sea — where the timeline boundary bulges hardest outward. When matter reaches the event horizon, it transitions from fully coupled to partially decoupled. Its quantum information is transferred to the TSC's permanent record, resolving the black hole information paradox.

Relativistic jets are the re-emergence of that energy as newly partially coupled flow line intersections ejected back into our timeline. For spinning (Kerr) black holes, frame-dragging organises the re-emergence into focused jets along the rotation axis. For non-spinning (Schwarzschild) black holes, re-emergence is more diffuse.

4. Dark Matter and Dark Energy

4.1 Dark Matter as Time-Decoupled Flow Lines

Dark matter is partially captured Timeless Sea material — flow lines that intersect our timeline at an angle that couples to gravity but not to time. Without time coupling there is no electromagnetic, weak, or strong interaction (these require time-coupled oscillations). Gravity acts on dark matter because gravity is TSC pressure, which acts on any flow line intersection regardless of time coupling.

UNIQUE PREDICTION	Atomic clocks placed near dark matter overdensities should show unexplained tiny time drifts. No other dark matter model predicts this signature.
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4.2 Dark Energy as Time Gradient

As galaxies move into the void toward the edge of expansion, they enter regions of lower matter density and therefore lower temporal coupling Γ . As Γ drops, the same TSC pressure ΔP produces higher perceived acceleration:

Observed Acceleration $\propto \Delta P / \Gamma(t)$

The speeding up of the universe is not a new force. It is because our timeline's grip on matter loosens as it expands into the void.

5. Flow Lines and Quantum Entanglement

5.1 Flow Lines Defined

A flow line is a one-dimensional, oriented, straight object existing in the Timeless Sea. It has no mass, charge, or spin — these emerge only when it intersects our timeline.

Our timeline is not static. It is a four-dimensional membrane sweeping through the Timeless Sea. Flow lines do not move; the timeline moves relative to them.

5.2 The Flipbook Analogy

Thread (flow line): static drawing across all pages

Pages (timeline): the moving sheet

Stickman (particle): intersection point seen at one frame

Measurement: creasing a page, forcing local full coupling ($\Gamma \rightarrow 1$)

The particle we observe is the creased page. The rest of the flow line remains in the TSC, which is why entanglement can persist after measurement of one endpoint.

5.3 Natural vs. Created Entanglement

Natural entanglement: The timeline sweeps across the same static flow line at two different locations. The two intersection points are naturally correlated — they are two points on the same object.

Created entanglement (SPDC): A laser cuts a flow line at a single point, creating two new free ends. The two photons are created simultaneously and are necessarily correlated — they were the same point a moment before.

5.4 Opposite Spins as Geometric Reflection

A flow line is oriented. When our timeline sweeps across it, the intersection occurs at two points: entry and exit. The oriented arrow is seen from opposite directions at these two points:

Entry → one spin orientation

Exit → opposite spin orientation

This is exactly analogous to a reflection in water. One object, two views, always opposite. No mystery — geometry.

5.5 The $\cos^2(\theta/2)$ Derivation

This section demonstrates interpretive compatibility rather than a full derivation from first principles.

The correlation formula follows from three assumptions:

A flow line has orientation angle φ , uniformly distributed over $[0, 2\pi)$ — randomness inherited from the expansion edge

A detector set to angle θ measures the projection of the flow line's orientation onto its axis

The projection rule for a spin- $\frac{1}{2}$ particle is amplitude = $\cos((\theta - \varphi)/2)$

The probability of detecting spin up is the square: $P(\text{up} \mid \theta, \varphi) = \cos^2((\theta - \varphi)/2)$. For two detectors measuring the two intersection points (φ and $\varphi + \pi$), the probability of opposite results integrates to $\cos^2((\theta_A - \theta_B)/2)$.

The framework does not replace quantum mechanics. It provides a geometric origin for the random phase φ and a mechanical interpretation of the singlet state.

5.6 Particle Stability and Capture

After decoherence, the ends of a cut flow line may become fully captured ($\Gamma = 1$), producing stable particles like electrons. Particles that remain partially coupled ($\Gamma < 1$) travel indefinitely (neutrinos) or decay when capture becomes unstable (muons, heavy quarks).

6. Qubit Coherence and Timeline Coupling

Qubit decoherence has two independent channels. The environmental coupling rate Γ_{meas} is the rate at which thermal photons, phonons, and electromagnetic noise measure the qubit. The **timeline coupling rate** Γ_{timeline} is the rate at which local matter density ρ increases temporal coupling between the qubit and our timeline.

$$Q_t = 1 / (\Gamma_{\text{timeline}} \times \Gamma_{\text{meas}})$$

Coherence times should improve monotonically with decreasing local matter density ρ (i.e., with increasing gravitational isolation), even when thermal and electromagnetic shielding are held constant.

6.1 The L2 Prediction

A superconducting qubit at the Sun–Earth L2 point should maintain coherence at an order-of-magnitude enhancement (potentially 10^2 scale under ideal conditions) compared to an identical qubit on Earth, with all thermal and electromagnetic shielding held constant.

Derivation	Result
$ \Phi_{\text{Earth}} / \Phi_{\text{L2}} $	$\approx 6.25 \times 10^7 / 2.6 \times 10^5 \approx 240\text{--}250$
If $\Gamma_{\text{timeline}} \propto \rho \propto \Phi $	$Q_t(\text{L2}) / Q_t(\text{Earth}) \approx 250$

Falsification threshold: if the measured coherence improvement at L2 is less than a factor of 50 (< 20% of the predicted 250×), the framework is falsified.

PART TWO: MATHEMATICAL GOALS

7. What Is Missing: The Mathematical Formalism

This framework does not yet have a complete mathematical formalism. The following are the critical-path derivations required for the framework to make fully quantitative predictions. They are listed here explicitly as an act of intellectual honesty and as a roadmap for future formalisation.

#	Goal	Required Derivation
P1	Timeline coupling	Derive $\Gamma(\rho) = \beta \rho^n$ from TSC dynamics, not observational fit. Confirm $n \approx 0.04$ theoretically.
P2	Pressure / Poisson	Show that $\nabla^2 P = 4\pi G \rho$ emerges from TSC dynamics, linking gravity-as-pressure to standard GR.
P3	Force law	Derive $g = -\nabla P$ as the Newtonian limit of TSC pressure gradient.

#	Goal	Required Derivation
P4	Planck threshold	Derive N_{critical} from coherence volume and TSC particle parameters.
P5	Relativistic limit	Show that TSC pressure reproduces the Einstein field equations in the relativistic limit.
P6	Galaxy formation	Derive $t_{\text{form}}(M,z) = t_{\text{LCDM}}(M,z) \times f(P_{\text{TSC}})$ for quantitative Webb predictions.
P7	Dark matter halos	Derive $\rho_{\text{DM}}(r) = \rho_{\text{TSC_capture}}(r) + \rho_{\text{wall}}(r)$ from TSC capture dynamics.
P8	$\cos^2(\theta/2)$	Derive the projection postulate from flow line geometry without assuming it.

PART THREE: OBSERVATIONAL CANDIDATES

The following sections list phenomena that are consistent with the framework and may be early signatures. None is claimed as proof. Each requires further testing and must rule out alternative explanations.

8. The Hubble Tension

The locally measured Hubble constant $H_{\text{local}} \approx 73 \text{ km/s/Mpc}$ differs from the CMB-derived value $H_{\text{CMB}} \approx 67 \text{ km/s/Mpc}$ by approximately 9%. In this framework that ratio is a direct measurement of the time dilation ratio between our local matter-dense region and the CMB surface:

$$T_{\text{local}} / T_{\text{CMB}} = H_{\text{local}} / H_{\text{CMB}} = 73/67 \approx 1.090$$

The CMB surface approaches TSC temporal conditions — very low matter density, very weak time coupling. The Hubble tension is not a measurement error. It is a candidate observational constraint on the time dilation gradient predicted by AETT.

9. The Webb Anomaly

JWST has observed massive, evolved galaxies at redshifts $z > 10$, with inferred stellar ages exceeding the cosmic timeline by factors of approximately 1.2 to 2.5. In this framework, this is consistent with the time rate power law. The proper time experienced by a protogalactic overdensity relative to the cosmic mean is:

$$\tau_{\text{dense}} / \tau_{\text{mean}} = (\rho_{\text{dense}} / \rho_{\text{mean}})^n$$

With $n \approx 0.04$ and overdensities $\delta = 100\text{--}1000$ in early protoclusters:

$$\delta = 100 \rightarrow \text{age excess factor} = 1.20 \text{ (20\% older)}$$

$$\delta = 1000 \rightarrow \text{age excess factor} = 1.32 \text{ (32\% older)}$$

The framework predicts that galaxy age excess should scale with local overdensity δ as $\delta^{0.04}$. Future JWST and ALMA observations can test this correlation directly.

10. Void Hubble Constant

The framework predicts that H_0 measured in cosmic voids should be measurably lower than in dense regions. With $n \approx 0.04$ and typical void density $\rho_{\text{void}} \approx 0.2 \rho_{\text{mean}}$:

$$H_{\text{void}} \approx 66 \pm 1.5 \text{ km/s/Mpc}$$

This is testable with existing DESI/Euclid data.

Falsification: if measured $H_{\text{void}} > 69$ or < 62 , or if the void–dense difference is $< 1\%$ (1σ), the framework is falsified.

11. Atomic Clock Dark Matter Signature

The framework uniquely predicts that atomic clocks placed near dark matter overdensities should show unexplained tiny time drifts. Because dark matter is time-decoupled, it perturbs local temporal density $\Gamma(\rho)$ without electromagnetic interaction.

This is testable with existing atomic clock technology and dark matter maps from gravitational lensing surveys. No other dark matter model predicts a time anomaly signature.

Falsification: if no correlation is found between atomic clock networks and dark matter overdensity maps with sufficient sensitivity, this unique prediction fails.

12. The Galactic Metabolism and the Fermi Bubbles

If the framework's account of continuous accumulation is correct, we should see signatures of **energy rejection at maximum contact points** — such as supermassive black holes — and signatures of **new matter entry in low-gravity null zones** such as cosmic voids.

The Fermi Bubbles — gamma-ray and X-ray lobes extending 25,000 light-years above and below the Galactic Centre — are one observed structure that may be consistent with the former. Their key properties:

Bipolar symmetry aligned with the Galactic rotation axis

Extended duration spanning millions of years

Hard gamma-ray spectrum not easily explained by standard processes

These properties are compatible with re-emergence from Sgr A* as a maximum contact point, with the outflow organised by the black hole's spin. Standard astrophysics has not reached consensus on the origin of the Fermi Bubbles (past Sgr A* outburst, starburst wind, dark matter annihilation). This framework offers a possible consistency — not a proof.

The framework predicts similar structures should be observable around other supermassive black holes in edge-on galaxies. Their absence in otherwise similar galaxies would challenge this interpretation.

Falsification: if a systematic survey of edge-on galaxies with similar central black holes finds no comparable bipolar structures, or if standard astrophysics produces a clean consensus explanation for the Fermi Bubbles, this candidate is ruled out.

13. Other Observational Candidates

The following two phenomena are mechanistically grounded in the framework. Each is given a full account here rather than a footnote, because both carry independent observational anchors.

13.1 Primordial Magnetic Fields

The same electrostatic aggregation process that drives early structure formation also generates primordial magnetic fields. Charged particles in collective motion produce magnetic fields. As particles aggregate electrostatically along preferred axes in the early universe, the magnetic fields they generate become aligned with those axes — which are the same axes that become the filaments of the cosmic web.

Filament geometry and field geometry share the same physical origin. This is why magnetic fields in the cosmic web align with filaments: they were born together from the same aggregation event, not one from the other.

$$\mathbf{B}(\mathbf{r}) = (\mu_0/4\pi) \int \mathbf{J}(\mathbf{r}') \times (\mathbf{r} - \mathbf{r}') / |\mathbf{r} - \mathbf{r}'|^3 d^3r'$$

This is testable via cross-correlation of magnetic field strength with galaxy overdensity at $z > 10$, combining Webb and Square Kilometre Array data. If fields show no alignment at these redshifts, or if alignment is fully explained by standard magnetohydrodynamics without an electrostatic aggregation component, this candidate is ruled out.

13.2 Matter-Antimatter Asymmetry

In April 2026, CERN announced a direct measurement of matter-antimatter asymmetry in a controlled experiment. A heavy, unstable neutron (a hyperon) was created by configuring quarks in a partial coupling regime. In 2.5% of 80,000 decays, the symmetry broke: a matter particle was created without its antimatter partner. Standard physics has no mechanism for this result.

Temporal Decoupling offers a specific explanation. When a flow line from the Timeless Sea intersects our timeline, it can couple in different modes. One mode produces matter; another produces antimatter; a third produces neutral dark matter. The expansion edge — the boundary where our timeline emerges from the Timeless Sea — is not perfectly neutral. A static charge difference between the timeline and the Timeless Sea favours one coupling mode over another. Small fluctuations at this boundary were amplified by the runaway coupling of the Big Bang, producing approximately one matter particle per billion antimatter partners.

The CERN experiment observed asymmetry precisely in a partial coupling regime — a particle not in fully coupled equilibrium. This is exactly what the framework predicts: asymmetry scales with deviation from full coupling. The more a particle is pushed away from $\Gamma = 1$, the more the boundary charge differential can bias its coupling mode.

PREDICTION	Asymmetry should increase as particles are pushed further from full coupling — into higher energy regimes or exotic partial-coupling configurations. Future accelerator experiments can test this directly.
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Falsification: if future accelerator experiments show no increase in matter-antimatter asymmetry as particles are pushed further from full coupling, the mechanism proposed here fails.

14. Final Statement

This framework is offered with humility and respect for the physics community. Two reinterpretations — gravity as pressure, time as emergent — connect more open problems than any single existing mechanism. The framework makes specific, falsifiable predictions. The most important is also the cheapest: qubit coherence times should improve measurably with gravitational isolation beyond what shielding alone predicts. That experiment can be run today, with existing hardware, for less than the cost of a modest research grant. If it fails, the framework fails cleanly. If it succeeds, everything else in this document becomes worth taking seriously.

Temporal Decoupling · Hollis Roberts · 2026 · Version 5

TSC · AETT

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Appendix A: Claims Register — Temporal Decoupling v5

This register lists every core claim made in this framework, its current status, the type of evidence required, and how it could be falsified. Claims are numbered for reference.

PART ONE: CORE MECHANICS

ID	Claim	Status	Evidence Required	Falsification
C1	The Timeless Sea (TSC) exists — a timeless medium outside our universe	Postulate	Indirect: successful predictions of derived claims (C2–C14)	If all derived predictions fail, the postulate is unnecessary
C2	Time emerges from matter density via $T(\rho) = \beta\rho^n$, with $n \approx 0.04 \pm 0.01$	Derived from observational fit	Compatible with Hubble tension, void H_0 , Webb age excess	If n is not consistently 0.04 across independent datasets
C3	Gravity is TSC pressure: $\nabla P = 4\pi G\rho$ — mathematically equivalent to GR	Postulate	Distinguishing test: qubit coherence vs. altitude (C10)	If qubit coherence does not improve with gravitational isolation
C4	Dark matter is time-decoupled flow lines — couples to gravity, not to time	Theoretical proposal	Null results from EM/weak DM detectors (already consistent); unique prediction (C5)	If DM is detected via EM or weak interaction
C5	Atomic clocks near dark matter overdensities show unexplained time drifts	Unique prediction	Correlation between atomic clock networks and DM lensing maps	If no correlation found with sufficient sensitivity
C6	Entanglement is two intersection points of the same static flow line	Interpretation	None beyond standard QM tests	Not falsifiable separately from QM
C7	Opposite spins are geometric reflection (entry vs. exit)	Derived from geometry	Compatible with QM correlations (already satisfied)	Not falsifiable separately from QM
C8	The $\cos^2(\theta/2)$ correlation follows from spinor projection + random ϕ	Derived (spin-1/2 projection)	Compatible with QM (already satisfied)	Not falsifiable separately from QM
C9	The arrow of time is increasing Γ_{coupling} : $dS/dt = k \cdot \Gamma$	Derived	Compatible with thermodynamics	If a mechanism for time asymmetry is found that does not involve coupling

PART TWO: MATHEMATICAL GOALS

ID	Claim	Status	Evidence Required	Falsification
M1	$\Gamma(\rho) = \beta\rho^n$ can be derived from TSC first principles ($n \approx 0.04$)	Not yet derived	Formal derivation	If derivation proves impossible or contradicts data
M2	$\nabla^2 P = 4\pi G\rho$ emerges from TSC dynamics	Not yet derived	Formal derivation	If derivation fails or contradicts GR
M3	Collective Planck threshold: $\Gamma_{\text{total}} \geq \Gamma_{\text{crit}}$ from coherence volume	Not yet formalised	Formal derivation from TSC condensate properties	If a rigorous no-go theorem shows collective threshold impossible

ID	Claim	Status	Evidence Required	Falsification
M4	TSC pressure reproduces the Einstein field equations in the relativistic limit	Not yet derived	Formal derivation	If derivation fails or contradicts GR

PART THREE: OBSERVATIONAL CANDIDATES

ID	Claim	Status	Evidence Required	Falsification
O1	The Hubble tension ($73/67 \approx 1.090$) is time dilation between local region and CMB surface	Derived from postulate + data	Confirmation that H_0 varies with environment as predicted (O2)	If environment-split H_0 shows no significant variation ($< 1\%$ difference)
O2	Environment-dependent Hubble constant: $H_{\text{void}} \approx 66 \pm 1.5$ km/s/Mpc	Quantitative prediction	DESI/Euclid environment-split BAO or cosmic chronometers	If $H_{\text{void}} > 69$ or < 62 , or void–dense difference $< 1\%$ (1σ)
O3	Webb galaxy age excess scales with overdensity δ as $\delta^{0.04}$ (15–32% older for $\delta = 100$ –1000)	Quantitative prediction	JWST/ALMA stellar population ages vs. environment density	If no correlation found, or scatter > 0.5 dex uncorrelated with δ
O4	Qubit coherence time improves with gravitational isolation; primary prediction: 250× longer at L2	Primary falsifiable test	Measured Q_t scaling from ground $\rightarrow$ LEO $\rightarrow$ GEO $\rightarrow$ L2	If L2 improvement $< 50\times$ ($< 20\%$ of predicted 250×)
O5	Dark matter will never be detected via EM or weak interactions (permanent null)	Unique prediction	Continued null results from liquid xenon etc., plus atomic clock test (C5)	If any EM/weak DM detection is confirmed
O6	The Fermi Bubbles are energy rejection at a galactic-scale maximum contact point (Sgr A*)	Observational candidate	Compatible with bipolar symmetry, duration, spectrum; similar structures in other galaxies	If another galaxy with similar central black hole shows no such structure; if standard physics produces a clean alternative explanation
O7	Similar bipolar structures exist around other supermassive black holes in edge-on galaxies	Predictive extension	X-ray/gamma-ray surveys of edge-on galaxies	If systematic survey finds none with similar properties
O8	Primordial magnetic fields aligned with cosmic web filaments arise from electrostatic aggregation	Observational candidate	Cross-correlation of field strength with galaxy overdensity at $z > 10$ (Webb + SKA)	If fields show no alignment, or alignment is fully explained by standard MHD
O9	Matter-antimatter asymmetry increases as particles are pushed further from full coupling	Observational candidate	Future accelerator experiments in higher-energy partial-coupling regimes; CERN 2026 consistent	If asymmetry does not increase with deviation from full coupling
O10	The LHC cannot test this framework — fully coupled particles ($\Gamma = 1$)	Boundary condition	Continued absence of anomalies in fully coupled collisions	If a clear, reproducible anomaly appears in fully coupled collisions unexplained by Standard Model

ID	Claim	Status	Evidence Required	Falsification
O11	The universe does not end in heat death — open system connected to TSC	Theoretical proposal	Detection of persistent low-level matter creation in voids/nebulae	If thermodynamic closure is proven (no low-entropy input)

Summary Table by Status

Status	Claim IDs
Postulate	C1, C3
Derived from data ($n \approx 0.04$)	C2
Derived from geometry / framework	C6, C7, C8, C9
Not yet derived — mathematical goal	M1, M2, M3, M4
Quantitative prediction (testable now)	O2 (H_void), O3 (Webb age), O4 (L2 qubit)
Unique prediction	C5 (atomic clock), O5 (permanent DM null)
Observational candidate	O6 (Fermi Bubbles), O8 (primordial B-fields), O9 (matter-antimatter)
Predictive extension	O7 (other galaxy bubbles)
Consistent with framework	O1, O11
Boundary condition	O10